

RCP Annualization

Setup

```
find_project_root <- function(start = getwd()) {
  path <- normalizePath(start, mustWork = FALSE)
  for (i in seq_len(8)) {
    if (dir.exists(file.path(path, "Code")) && dir.exists(file.path(path, "Data"))) return(path)
    parent <- dirname(path)
    if (identical(parent, path)) break
    path <- parent
  }
  stop("Set LC_DLNM_BASE_DIR to the project root.")
}

base_dir <- Sys.getenv("LC_DLNM_BASE_DIR")
if (!nzchar(base_dir)) base_dir <- find_project_root()
default_params <- list(run_annualization = TRUE, group_size = 52)
rmd_params <- if (exists("params") && !is.null(params)) {
  modifyList(default_params, as.list(params))
} else default_params

invisible(lapply(list.files(file.path(base_dir, "Code/Function"),
                             pattern = "\\R$", full.names = TRUE), source))

simulation_dir <- file.path(base_dir, "Results/Simulated_mortality_rates")
annualized_dir <- file.path(base_dir, "Results/Simulated_annualized_mortality_rates")
dir.create(annualized_dir, recursive = TRUE, showWarnings = FALSE)

model_names <- c("LC", "LL")
scenario_names <- c("rcp26", "rcp85")
```

Annualize Weekly Simulation Results

```
annualized_results <- list()

for (model in model_names) {
  annualized_results[[model]] <- list()

  for (scenario in scenario_names) {
    weekly_simulation <- load_rcp_weekly_simulation(
      model = model,
      scenario = scenario,
      simulation_dir = simulation_dir
    )
    annualized_results[[model]][[scenario]] <- annualize_rcp_simulation(
      weekly_simulation,
      group_size = rmd_params$group_size
    )
  }
}
```

```
)  
  
save_rcp_annualized_simulation(  
    annualized_results[[model]][[scenario]],  
    model = model,  
    scenario = scenario,  
    annualized_dir = annualized_dir  
)  
}  
}
```